Aditya Velpula

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WORK EXPERIENCE

Graduate Teaching Assistant **Aug 2025 - Present**
George Mason University ***Fairfax, VA***

- Handpicked by Professor Harry J. Foxwell to support 2 graduate sections in AIT-580, improving assignment clarity and reducing regrade requests.
- Mentored 50+ graduate students on Python, SQL, R, and AWS Cloud assignments, resulting in measurable improvements in project accuracy and on-time submissions.
- Helped students with projects that used cloud-based analytics pipelines, data modeling, and SQL queries.
- Collaborated with faculty to enhance course materials and streamline assessment workflows through data-driven insights.

AI Engineer **Nov 2023 - Jul 2024**
Indgeos Geospatial ***Telangana, India***

- Developed Python algorithms for large-scale geospatial image processing, improving pipeline performance.
- Established data-quality checks and feature engineering methods to improve ML readiness.
- Collaborated with version control (Git) and testing to ensure reliable data flow; designed and built the company website.

PROJECTS

Obesity Risk Analytics Using AWS Cloud Infrastructure (Under the Supervision of Prof. Harry Foxwell)

- Created an end-to-end data pipeline using AWS S3, RDS (MySQL), and Glue to ingest, clean, and profile CDC public health datasets.
- Performed SQL-based statistical analysis to study relationships between obesity, lifestyle behaviors, and socio-economic indicators.
- Built Python/R visualizations using Dash, Seaborn, and ggplot2 to communicate regional insights.

U.S. Electricity-Rate Analytics: DIKW-Driven Exploration of IOU vs. Non-IOU Pricing

- Analyzed 320K+ electricity-rate records (2020–2023) using Python and SQL to compare pricing patterns between IOU and non-IOU utilities.
- Applied regression and clustering techniques to identify geographic and sector-based pricing differences.

Wildfire Risk Prediction Using Climate Data & Machine Learning

- Integrated large-scale satellite and climate datasets (MODIS, NOAA) to engineer geospatial and temporal features for wildfire risk modeling.
- Automated preprocessing workflows and trained Random Forest and XGBoost models, achieving an AUC of 0.99 for high-accuracy risk classification.

SKILLS

- **Languages:** Python, C/C++, Java, MATLAB (Beginner), SQL, R, Software Engineering: Algorithms, OOP, Testing, Debugging, Version Control.
- **Cloud Platforms:** AWS (S3, RDS, Glue), Azure (Data Factory, Databricks), Basic GCP.
- **Data Engineering:** ETL/ELT Pipelines, Data Modeling, Data Warehousing, Data Validation, Data Pipelines.
- **Libraries:** Pandas, NumPy, PySpark, Seaborn, Matplotlib, Scikit-learn.
- **Tools:** Git, Jupyter, Linux CLI, Power BI, Spark (Basics), Airflow (Basics).
- **Concepts:** LLMs, Vector Databases, RAG Concepts, Prompt Engineering (Basic), BI, Data Analysis, Data Wrangling, EDA, Regression, Classification, Feature Engineering, Model Evaluation.

EDUCATION

Masters, Data Analytics ***2024-2026***
George Mason University, Fairfax, Virginia

B. Tech, Computer Science and Engineering (AI&ML) ***2020-2024***
St. Peters Engineering College, Telangana, India

CERTIFICATIONS

- **AWS Certified Generative AI Developer – Professional (Early Adopter) (Verification)**
- **AWS Certified Machine Learning Engineer – Associate (Verification)**
- **AWS Certified Data Engineer – Associate (Verification)**
- **AWS Certified Solutions Architect – Associate (Verification)**
- **AWS Certified AI Practitioner – Foundational (Verification)**
- **AWS Academy Graduate - Data Engineering - Training Badge (Verification)**
- **AWS Academy Graduate - Cloud Foundations - Training Badge (Verification)**
- **AI For Everyone (Verification)**
- **SQL (Advanced) Certificate (Verification)**